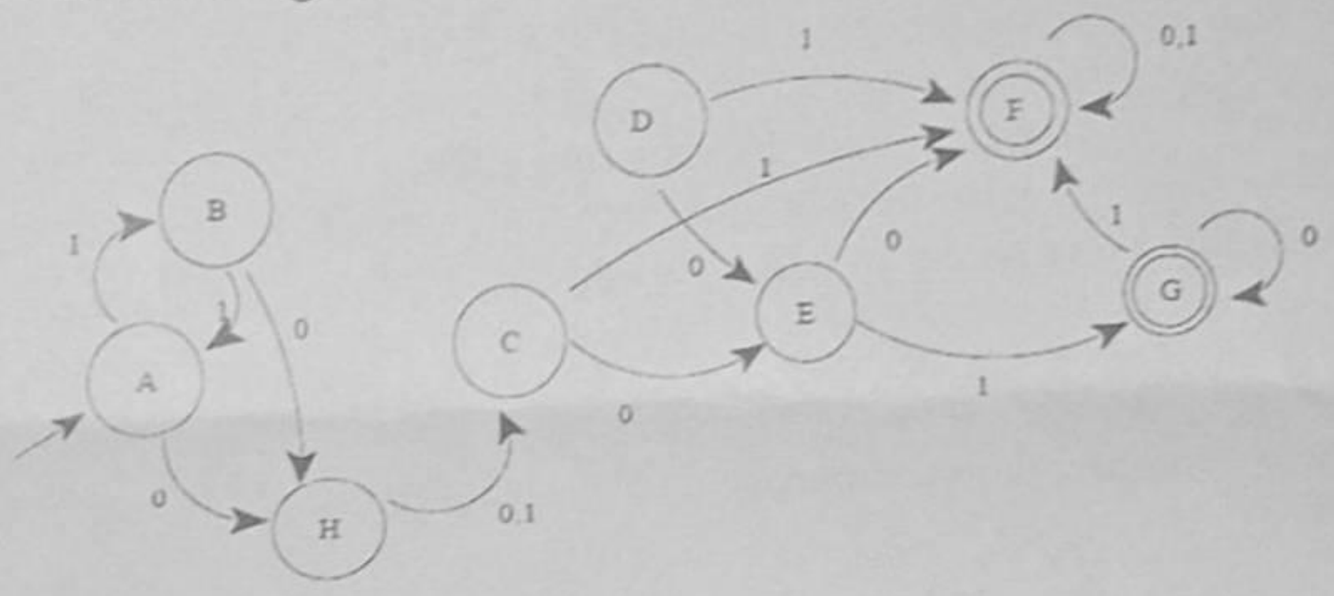
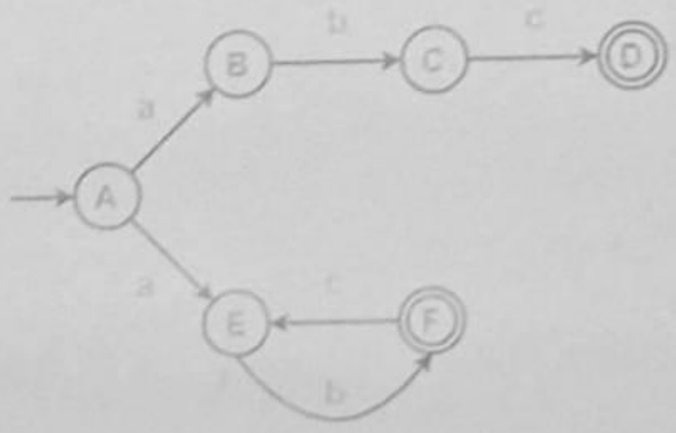
B.Tech. Comp. Engg. Vth Sem Examination 2023
AUTOMATA THEORY
Paper no CEN-502

Max Time : Three hours | Max. Marks: 60

Attempt any TWO parts from each question.

Write your Roll No here ..... on receipt of the question paper immediately

Q. No	CO	Questions	Marks
Q1 a)	1	Explain the differences between NFA and DFA using suitable examples.	6
Q1 b)	1	Design a relevant machine for the following and validate your answers: i) DFA for language **(a*+b*) (b + ab)* c*** ii) A Mealy machine over {0, 1} which outputs EVEN/ODD according to the number of 1's observed in the input. iii) A Moore Machine which converts each occurrence of substring **100** in the input by a **101** .	6
Q1 c)	1	Minimize the following DFA ***with the Matrix Method only***.  **Verify your answers using suitable sentences.**	6
Q2 a)	2	State the 13 identities generally used for simplification of Regular Expressions.	6
Q2 b)	2	Construct the Regular Expression for the given Automaton showing all intermediate steps.  ***Verify your answers using suitable sentences.***	6

Q2 c)	2	State and Define the Pumping Lemma. Also comment on the Regularity of the following language using this lemma i) $L1 = \{ww \quad : w \in \{a, b\}^*\}$	6
Q3 a)	2	Explain the concept of Ambiguity in Grammars using suitable examples. Also explain the Chomsky classification of Languages in details.	6
Q3 b)	3	Construct a CFG for a language described as follows : $L_1 = \{ (a + b)^n b\ (x + xy)^n$, **where n > 1**$\}$ **Verify your answers using suitable sentences.**	6
Q3 c)	3	Simplify the following CFG. $S \rightarrow ABaC \mid Ba$ $\quad B \rightarrow bB \mid b$ $A \rightarrow aC \mid CC$ $\quad C \rightarrow d \mid \varepsilon$ **Mention all the steps and the corresponding rules used at appropriate positions.**	6
Q4 a)	4	Mathematically define a Push Down Automaton. Explain how the structure of a PDA alleviates the challenges faced by FA.	6
Q4 b)	4	Construct a Deterministic PDA for the followings: i) Language L_1 which accepts set of all strings over {0, 1} which do not contain the substring 101. ii) $L_2 = \{ a^n b^m c^{n+m}$: such that n, m >=0$\}$. **Validate your answers using suitable examples**	6
Q4 c)	4	Convert the following CFG to PDA $X \rightarrow XYZ \mid XxYyZz$ $\quad Y \rightarrow Yy \mid XZ$ $\quad Z \rightarrow z$ $X \rightarrow xy$ $\quad Y \rightarrow \varepsilon$ Also using the tabular approach, check whether the string is acceptable by this PDA or not: **xxyyz**	6
Q5 a)	5	Construct a Turing Machine for the followings operations: **String Reversed Copy** : Given a string on the tape , the TM copies the reverse of the original string on the tape after one blank on the right of the original string; ***Verify your machine using suitable example.***	6
Q5 b)	5	Prove that the function $f5(a, b, c) = (a^2b) - (a^5)/c$ is primitive recursive. *Assume the basic math operations like add (f1), multiply (f2), subtract (f3) divide(f4) to be primitive recursive.*	6
Q5 c)	5	Explain following terms in details with suitable examples: i) Church's Thesis ii) NP Complete Problems iii) Recursive Languages	6

**********End of Paper**********